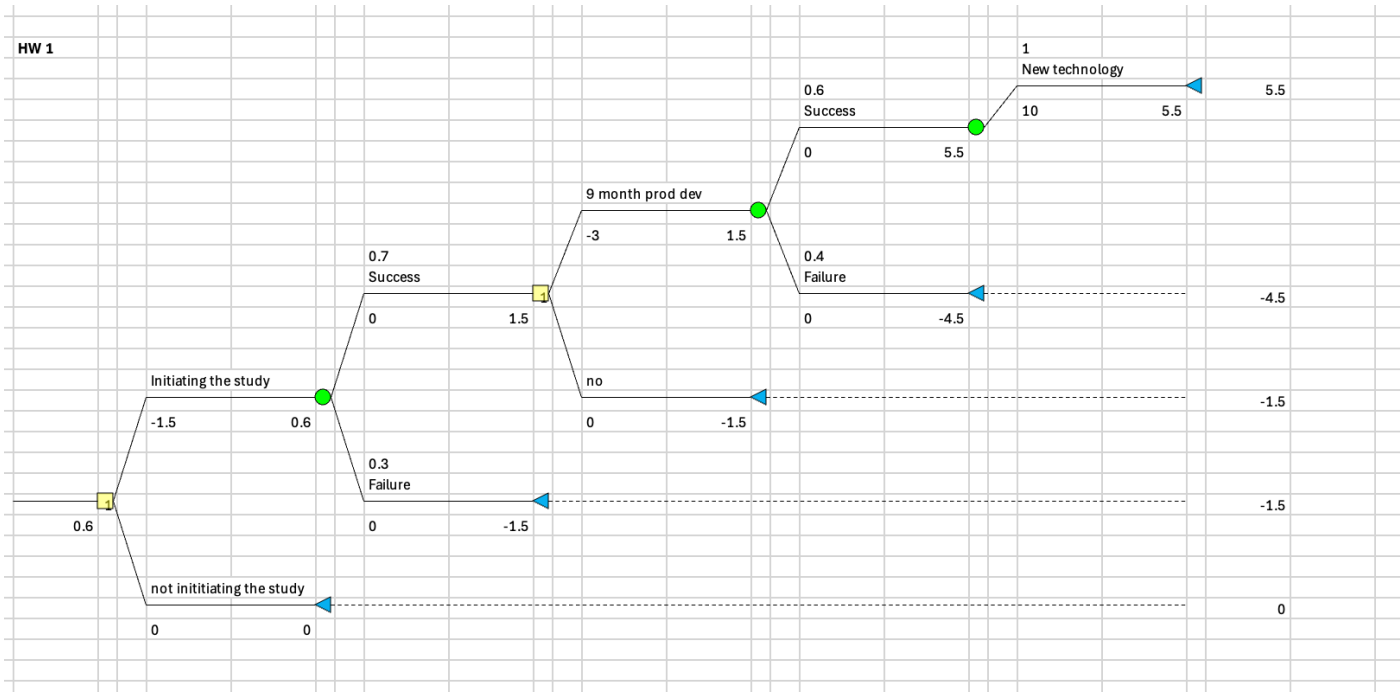
SHWETA MITTAL

Decision Models

HW case-1

Part 1:

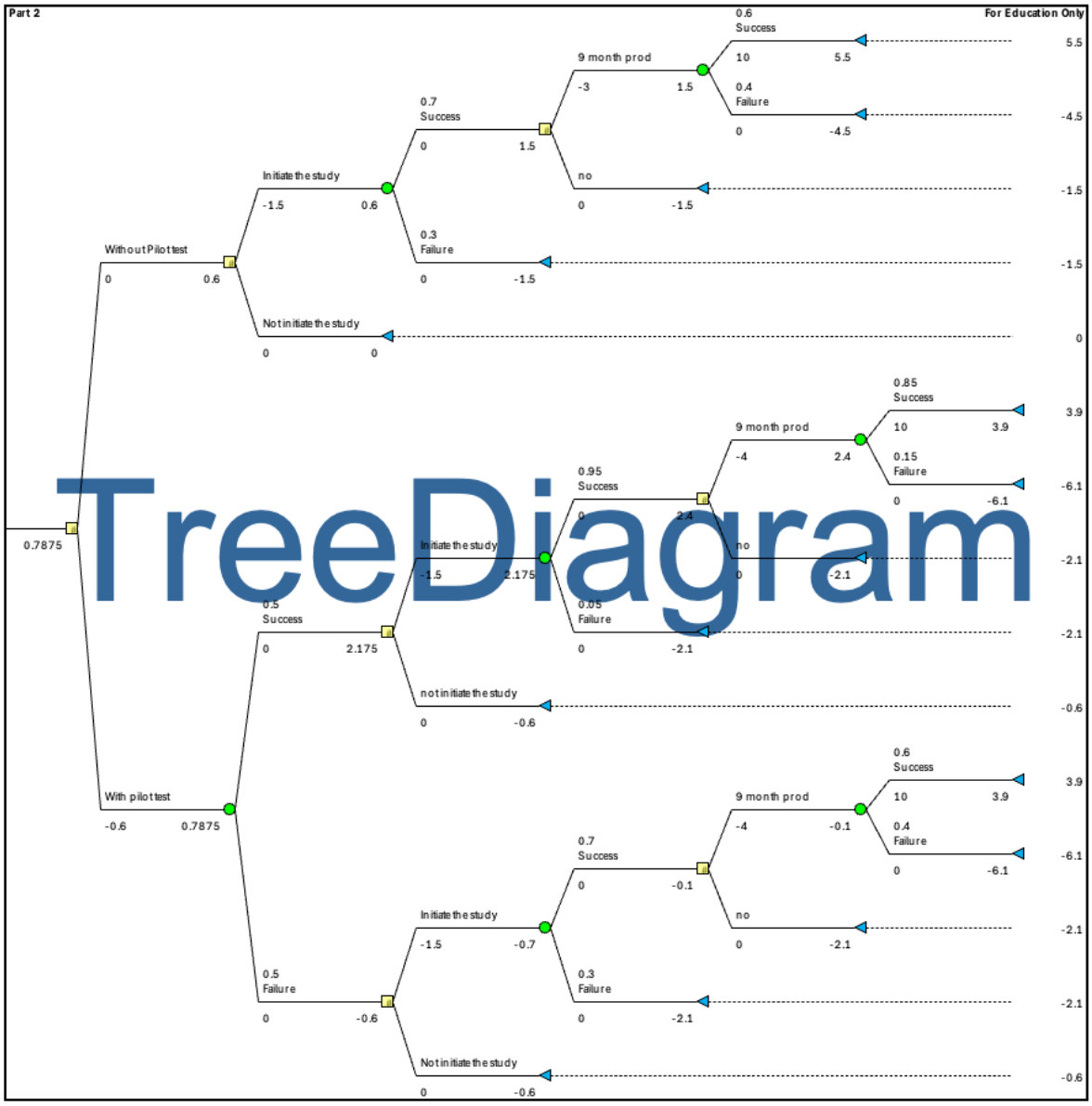


The Firm **should initiate the project** since the expected profit will be 0.6 million dollars if the study is initiated.

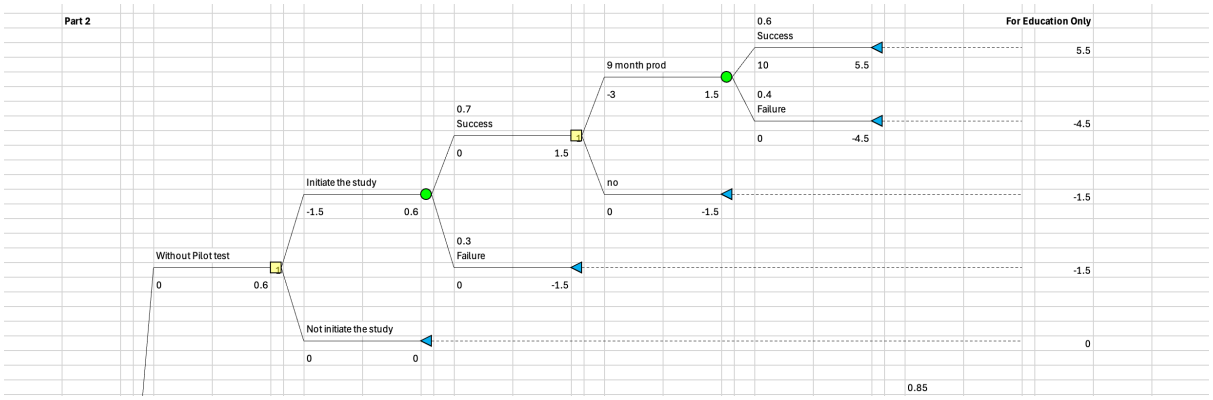
Condition under which it should continue the development if the feasibility study is successful which will generate an expected profit of 1.5 million dollars.

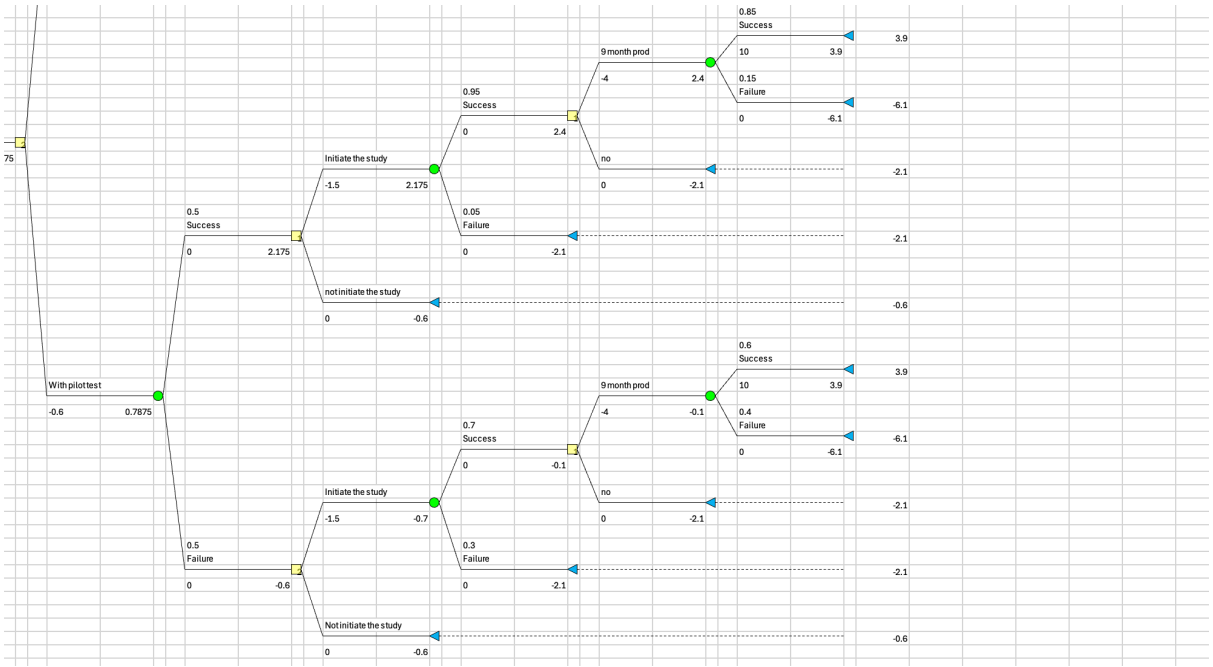
However, it should abandon the development if feasibility study fails which will led to a loss of 1.5 million dollars.

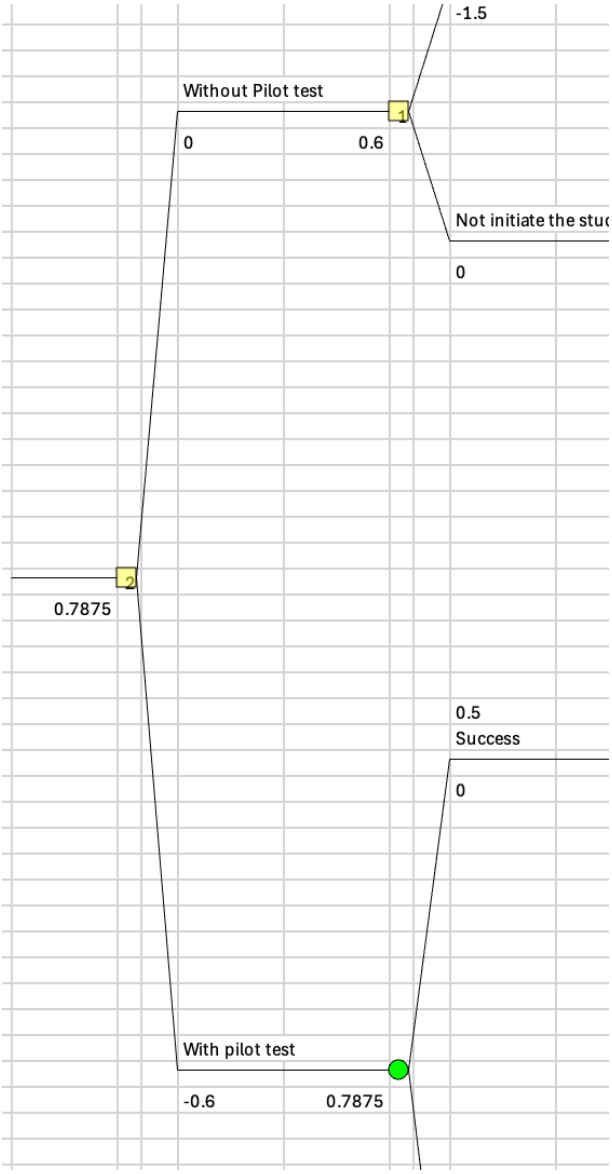
PART2:



TreeDiagram


Part 2
For Education Only
0.6
Success
9 month prod
10
5.5
-3
1.5
0.4
Failure
0
-4.5
0.7
Success
0
1.5
Initiate the study
-1.5
0.6
no
0
-1.5
0.3
Failure
0
-1.5
Without Pilot test
0
0.6
Not initiate the study
0
0
0.85
5.5
-4.5
-1.5
-1.5
0


75
0.85
Success
9 month prod
10
3.9
-4
2.4
0.15
Failure
0
-6.1
0.95
Success
0
2.4
Initiate the study
-1.5
2.175
no
0
-2.1
0.05
Failure
0
-2.1
0.5
Success
0
2.175
not initiate the study
0
-0.6
With pilot test
-0.6
0.7875
0.6
Success
9 month prod
10
3.9
-4
-0.1
0.4
Failure
0
-6.1
0.7
Success
0
-0.1
Initiate the study
-1.5
-0.7
no
0
-2.1
0.5
Failure
0
-0.6
0.3
Failure
0
-2.1
Not initiate the study
0
-0.6
3.9
-6.1
-2.1
-2.1
-0.6
3.9
-6.1
-2.1
-2.1
-0.6



2. The firm should conduct the pilot testing since expected profit would be 0.7875 million dollars. If pilot test succeeds, the firm should initiate the study since expected profit would be 2.175 million dollar. However, if the pilot test fails, the firm should not initiate the study since the expected loss would be 0.7 million dollars from initiating the study.

3. The optimal net profit with pilot test would be 0.7875 million dollars . The optimal net profit without pilot test would be 0.6 million dollars.